Wren Fairbourne

Algebra 2

Graded Sample Quiz

Unit 3, Lesson 3.1

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ [B]

2. $x^2 \cdot x^4$ [C]

3. $(2x-1)-(x-4)$ [F]

4. $(x-3)(x+3)$ [G]

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

$5x^3-2x^3\ x^2+3x^2-2x^3+3x^2-x-6$

[$3x^3+4x^2-5x-4$]

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

[2]

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

[6, 2]

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 = 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

She forgot to distribute the negative to all the terms. $3x^2+2x-5-x^2+4x-1$

Roz Ingleby

Algebra 2

Graded Sample Quiz

Unit 3, Lesson 3.1

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	B. x^2+9	C. x^6	D. x^8	E. $x+3$	F. $x-5$	G. x^2-9

1. $(x+3)^2$ [B]

2. $x^2 \cdot x^4$ [C]

3. $(2x-1)-(x-4)$ [F]

4. $(x-3)(x+3)$ [G]

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

$3x^3 + 4x^2 - 3x + 8$

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

2

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

3, 5

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 \;=\; 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

The mistake is in the first step. She only subtracted the x^2 and forgot that the $-4x$ and the $+1$ change signs too. Answer: $2x^2 + 6x - 6$

Anouk Kittredge-Moss

Algebra 2

Graded Sample Quiz

Unit 3, Lesson 3.1

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ [B]

2. $x^2 \cdot x^4$ [C]

3. $(2x-1)-(x-4)$ [E]

4. $(x-3)(x+3)$ [G]

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

$3x^3 + 4x^2 - 5x - 4$

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

2

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

6, 3

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 \;=\; 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

Her first step is where it went wrong.

Algebra 2

Graded Sample Quiz

Unit 3, Lesson 3.1

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ — A

2. $x^2 \cdot x^4$ — D

3. $(2x-1)-(x-4)$ — E

4. $(x-3)(x+3)$ — G

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

$3x^3 + 4x^2 - 5x - 4$

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

3/2

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

3, 5

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 \;=\; 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

She should have added them instead, $4x^2 - 2x - 4$

Bina Galloway-Prue

Algebra 2

Graded Sample Quiz

Unit 3, Lesson 3.1

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ [B]

2. $x^2 \cdot x^4$ [D]

3. $(2x-1)-(x-4)$ [E]

4. $(x-3)(x+3)$ [G]

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

[ ]

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

[14]

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

[6, 2]

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 = 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

[Her first line is wrong. The answer should be $2x^2+6x-6$]

Tamsin Bellweather

Algebra 2

Graded Sample Quiz

Unit 3, Lesson 3.1

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ — B

2. $x^2 \cdot x^4$ — C

3. $(2x-1)-(x-4)$ — F

4. $(x-3)(x+3)$ — B

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

   $(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

   $3x^3 + 4x^2 - 3x + 8$

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

   3/2

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

   5, 3

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 \;=\; 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

Her first step is wrong: the minus goes to all three terms, so $-(x^2 - 4x + 1) = -x^2 + 4x - 1$. Then $3x^2 + 2x - 5 - x^2 + 4x - 1 = 2x^2 + 6x - 6$

Imogen Delacorte-Vey

Algebra 2 — Unit 3, Lesson 3.1

Graded Sample Quiz

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ — B

2. $x^2 \cdot x^4$ — D

3. $(2x-1)-(x-4)$ — F

4. $(x-3)(x+3)$ — B

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

$5x^3-2x^3\ x^2-3x^2\ -2x^3\ -3x^2\ +x\ +6$

$3x^3 - 2x^2 - 3x + 8$

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

3/2

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

6, 2

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 \;=\; 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

The mistake is in the first step. She only subtracted the x^2 and forgot that the $-4x$ and the $+1$ change signs too. Answer: $2x^2 + 6x - 6$

Liesl Halvorsen-Teague

Algebra 2

Graded Sample Quiz

Unit 3, Lesson 3.1

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ [A]

2. $x^2 \cdot x^4$ [D]

3. $(2x-1)-(x-4)$ [F]

4. $(x-3)(x+3)$ [B]

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

   $(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

   $3x^3 + 4x^2 - 5x - 4$

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

   14

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

   3, 5

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 \;=\; 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

The first line is wrong. It should say $3x^2 + 2x - 5 - x^2 + 4x - 1$, which makes $2x^2 + 6x - 6$

Detta Castellane

Algebra 2
Graded Sample Quiz

Unit 3, Lesson 3.1

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching *1 point each*

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ — B

2. $x^2 \cdot x^4$ — D

3. $(2x-1)-(x-4)$ — E

4. $(x-3)(x+3)$ — B

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*
$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

$5x^3-2x^3\ x^2+3x^2\ -2x^3\ +3x^2\ -x\ -6$

$3x^3 + 4x^2 - 5x - 4$

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

2

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

3, 5

Part C — Explain a step *4 points*

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 \;=\; 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

She messed up the signs. The answer is $2x^2 + 6x - 6$

Coralie Jessamyn

Algebra 2 — Unit 3, Lesson 3.1
Graded Sample Quiz

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ [A]

2. $x^2 \cdot x^4$ [C]

3. $(2x-1)-(x-4)$ [E]

4. $(x-3)(x+3)$ [B]

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*
$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

[$3x^3 + 4x^2 - 3x + 8$]

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

[1.5]

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

[5, 3]

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 = 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

You have to subtract every term, not just the first one, so the answer is $2x^2 + 6x - 6$

June Arden

Algebra 2 — Unit 3, Lesson 3.1

Graded Sample Quiz

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	B. x^2+9	C. x^6	D. x^8	E. $x+3$	F. $x-5$	G. x^2-9

1. $(x+3)^2$ — A

2. $x^2 \cdot x^4$ — D

3. $(2x-1)-(x-4)$ — E

4. $(x-3)(x+3)$ — G

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

$5x^3-2x^3\ x^2-3x^2\ -2x^3\ -3x^2\ +x\ +6$

$3x^3 - 2x^2 - 3x + 8$

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

3/2

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

6, 3

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 \;=\; 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

She went wrong when she took off the parentheses. Subtracting means every term of $x^2 - 4x + 1$ changes sign, not just the x^2. It should be $3x^2 + 2x - 5 - x^2 + 4x - 1 = 2x^2 + 6x - 6$

Marisol Ellisworth

Algebra 2

Graded Sample Quiz

Unit 3, Lesson 3.1

No notes. Write each final answer **inside its box**; show your work in the space around it. Only what is inside a box is graded.

Part A — Matching

1 point each

Match each expression with its simplified form. Write the letter in the box. Not every letter is used.

A. x^2+6x+9	**B.** x^2+9	**C.** x^6	**D.** x^8	**E.** $x+3$	**F.** $x-5$	**G.** x^2-9

1. $(x+3)^2$ — A

2. $x^2 \cdot x^4$ — C

3. $(2x-1)-(x-4)$ — F

4. $(x-3)(x+3)$ — G

Part B — Short answer

5. **Subtract,** and write the result in standard form. *(2 points)*

$(5x^3+x^2-4x+2)-(2x^3-3x^2+x+6)$

$3x^3 + 4x^2 - 5x - 4$

6. **Evaluate** $P(x)=8x^3+2x^2$ at $x=\frac{1}{2}$. *(1 point)*

3/2

7. **Expand** $x^2(x^3+x)$. List *every* exponent of x in your answer, separated by commas. *(2 points)*

6, 3

Part C — Explain a step

4 points

8. Maya simplified $(3x^2+2x-5)-(x^2-4x+1)$ like this:

$$3x^2+2x-5-x^2-4x+1 \;=\; 2x^2-2x-4$$

Her answer is wrong. Name the step where she went wrong, explain what she should have done there, and give the correct answer.

Her answer is right because you combine like terms: $3x^2 - x^2 = 2x^2$ and $2x - 4x = -2x$